Executive Summary

Tosco Branch Re-Entry Wells — Production Potential, Geology & Drilling Execution

Bowman County, North Dakota · Williston Basin southern margin (Red River trend)

Prepared for: Marlo Operating Company · Valor Energy Partners | Wells: Branch 4 RR (#41700) · Branch 2 (#41258)

Bottom line

Both wells are RE-ENTRIES into a pool Marlo just discovered and is already producing (sister well Branch 5). That makes them DEVELOPMENT-CLASS step-outs, not wildcats: the reservoir is effectively proven, and the outcome turns on EXECUTION of the short-radius re-entry rather than on whether oil is present. We assess a ~64–72% probability of a successful producer (modest size), with the dominant risk being mechanical — and largely knowable in advance from the original wellbore's logs.

~64–72%
Probability of a successful producer

Development-class re-entry
Not a wildcat (pool is producing)

Execution
Dominant risk: short-radius mechanics

~65–150 bopd
IP if successful · EUR 40–150 Mbbl (direct offset Branch 5 ≈ 65 bopd)

Chance of success — development-class, well above the ~20% wildcat line

RECOMMENDED ACTION

Pull the original Branch 4 wellbore's casing / cement-bond log and confirm the target zone. That one step sets the dominant (mechanical) risk and collapses most of the 40–85% band toward the high end before committing the curve.

EXECUTION RISK — TRACKED IN THREE LAYERS

Mechanical
Front-loaded — kickoff & casing integrity, knowable in advance from the original wellbore logs.

Geological
Landing and holding the thin 'B'; the offset / fleet dip prior keeps the lateral in zone.

Fluid — loss & gain
Mud lost to natural fractures; water gain at the OWC — held with MPD / LCM / ECD discipline.

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